

YUBILEYNAYA, Russian Federation

WMO: 219310

Lat: 70.767N

Lon: 136.217E

Elev: 24

StdP: 101.03

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-45.9	-44.2	-47.6	0.0	-45.6	-46.1	0.0	-44.1	11.0	-28.6	9.1	-21.6	1.9	210	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.1	24.7	16.0	22.0	14.8	19.5	13.5	17.3	22.9	15.6	20.7	14.1	18.7	3.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.9	10.6	19.5	13.2	9.5	17.8	11.6	8.5	16.5	48.9	22.8	44.1	20.8	39.6	18.8	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
10.4	8.7	7.3	DB	-47.1	28.7	1.6	2.4	-48.2	30.4	-49.2	31.8	-50.0	33.2	-51.2	34.9	
			WB	-47.1	19.0	1.6	1.4	-48.2	20.0	-49.2	20.8	-50.0	21.5	-51.2	22.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-11.8	-33.9	-33.2	-24.7	-14.4	-3.6	8.4	11.4	10.5	2.9	-10.7	-23.5	-32.2
	DBStd	17.92	6.50	6.87	8.11	6.87	6.28	5.55	5.03	3.76	3.68	6.98	6.61	6.34
	HDD10.0	8149	1361	1209	1077	733	423	95	43	39	216	641	1004	1308
	HDD18.3	11011	1619	1442	1335	983	680	301	223	244	463	899	1254	1567
	CDD10.0	192	0	0	0	0	1	48	86	54	2	0	0	0
	CDD18.3	13	0	0	0	0	0	4	8	1	0	0	0	0
	CDH23.3	136	0	0	0	0	0	43	82	11	0	0	0	0
	CDH26.7	28	0	0	0	0	0	6	22	0	0	0	0	0
Wind	WSAvg	3.1	2.5	2.3	2.6	2.8	3.5	4.1	3.8	3.4	3.0	3.0	3.3	2.6
Precipitation	PrecAvg	245	8	9	8	8	17	28	53	41	27	22	14	11
	PrecMax	454	14	30	15	15	36	63	125	116	53	114	29	28
	PrecMin	183	2	1	2	4	2	1	14	16	2	2	2	2
	PrecStd	63	4	7	3	3	10	16	34	23	13	24	8	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-15.9	-12.1	-2.6	4.3	17.1	27.6	29.2	24.9	17.3	1.0	-5.8	-14.3
		MCWB	-16.5	-12.9	-3.9	0.5	9.9	16.5	18.4	15.7	10.2	-0.9	-6.8	-14.8
	2%	DB	-18.2	-15.8	-5.3	1.5	11.8	24.2	25.4	22.2	13.6	0.0	-8.5	-17.4
		MCWB	-18.5	-16.3	-6.7	-1.4	6.5	14.8	16.9	14.8	8.3	-1.1	-8.9	-17.8
	5%	DB	-20.9	-20.3	-8.4	-1.0	7.8	21.1	22.4	19.7	10.6	-0.8	-11.1	-20.1
		MCWB	-21.1	-20.6	-9.4	-3.5	3.9	13.5	15.6	13.9	7.3	-1.9	-11.7	-20.4
	10%	DB	-24.2	-23.9	-12.1	-3.8	5.1	17.9	19.8	17.3	8.2	-1.9	-13.5	-22.5
		MCWB	-24.4	-24.2	-13.1	-5.7	2.4	12.0	14.1	13.0	5.8	-2.9	-13.9	-22.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-16.3	-13.0	-3.8	0.8	10.2	17.5	19.6	18.0	10.7	0.0	-6.6	-14.7
		MCDB	-16.0	-12.1	-2.9	3.7	16.6	25.1	26.3	22.0	15.6	0.6	-5.9	-14.3
	2%	WB	-18.5	-16.3	-6.7	-1.2	6.7	15.9	18.0	16.3	9.2	-0.9	-9.0	-17.7
		MCDB	-18.3	-15.8	-5.6	1.6	11.4	22.8	23.7	20.4	12.8	-0.3	-8.5	-17.4
	5%	WB	-21.1	-20.5	-9.4	-3.4	4.2	14.0	16.3	14.8	7.5	-1.8	-11.7	-20.3
		MCDB	-20.9	-20.3	-8.3	-1.1	7.2	20.6	21.6	18.3	9.9	-0.9	-11.1	-20.0
	10%	WB	-24.5	-24.1	-13.1	-5.7	2.5	12.1	14.5	13.3	6.0	-2.9	-13.9	-22.7
		MCDB	-24.3	-23.9	-12.2	-3.9	4.9	17.4	19.0	16.6	8.3	-1.9	-13.5	-22.5
Mean Daily Temperature Range	5% DB	MDBR	5.7	6.2	8.4	9.6	7.0	8.4	8.1	7.7	5.4	5.2	6.0	5.7
		MCDBR	7.1	9.6	9.3	11.5	10.7	13.4	12.4	12.4	9.7	4.1	7.6	7.3
		MCWBR	6.9	9.3	8.6	9.2	6.0	6.6	5.8	6.4	5.5	3.7	7.5	7.2
	5% WB	MCDBR	7.1	9.5	9.3	11.4	9.9	12.4	11.7	10.8	8.9	3.7	7.6	7.3
		MCWBR	6.8	9.3	8.5	9.2	5.6	6.5	6.3	6.5	5.5	3.5	7.3	7.1
Clear-Sky Solar Irradiance	taub	N/A	0.203	0.279	0.317	0.354	0.352	0.408	0.375	0.297	0.217	N/A	N/A	
	taud	N/A	2.004	2.062	2.062	2.167	2.420	2.263	2.339	2.466	2.082	N/A	N/A	
	Ebn at Noon	0	501	695	793	812	836	758	739	717	459	0	0	
	Edh at Noon	0	36	79	112	116	94	107	88	56	32	0	0	
All-Sky Solar Radiation	RadAvg	0.00	0.32	1.89	4.37	5.96	5.55	4.73	3.21	1.60	0.54	0.00	0.00	
	RadStd	0.00	0.03	0.08	0.14	0.32	0.47	0.31	0.36	0.12	0.07	0.00	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page